

Operating Period

Annual	350 days
Daily	16 hours
Hours Annually	5600 hours
Planned Outage	10 days
Unplanned Outage	5 days

Production Rate

Annually	1,000,000 gallons/year
Daily	2,857 gallons/day
Hourly	179 gallons/hour

Component Data

Yellow Grease

Moisture	0.5% by Weight
FFA	5.0% by Weight
Other	1.5% by Weight
Temperature	120 °F
Specific Heat Capacity	0.420 BTU/(lbm-°F)
Density	0.894 g/cc
TG Avg. Mol. Mass	868.66 g/mol.
FFA Avg. Mol. Mass	276.87 g/mol.

Methanol

Density	0.791 g/cc
Mol. Mass	32.04 g/mol.
Temperature	55 °F
Specific Heat Capacity	0.623 BTU/(lbm-°F)
Latent Heat of Vaporization	424 BTU/lbm

Catalyst (KOH)

Density	2.044 g/cc
Mol. Mass	56.11 g/mol.

Biodiesel

Avg. Mol. Mass	290.90 g/mol.
Density	0.874 g/cc
Specific Heat Capacity	0.537 BTU/(lbm-°F)

Glycerol

Avg. Mol. Mass	92.09 g/mol.
Density	1.261 g/cc
Specific Heat Capacity	0.661 BTU/(lbm-°F)

Citric Acid

Concentration	100%
Avg. Mol. Mass	192.12 g/mol.

Density	1.665 g/cc
Water	
Avg. Mol. Mass	18.02 g/mol.
Density	1.000 g/cc
Latent Heat of Vaporization	970 BTU/lbm
Temperature	55 °F
Specific Heat Capacity	0.989 BTU/(lbm-°F)
Process Conditions	
Pre-Heat Tank	
Material	Steel
Pre-Heat Temperature	140 °F
Volume	2,100 gallons
UA of Heating Coil	9,600 BTU/(hr-°F)
Reactor	
Material	Steel
Volume	900 gallons
Reaction Yield	99.7% by Weight
Heat Loss	2.5%
Pelly A5	
Volume	80 gallons
Catalyst	KOH
Catalyst Dosage	0.40% VO by Weight
Methanol Dosage	20.0% by Volume
	17.7% by Weight
	4.800 Molar Ratio to VO
Reaction Time	180 minutes
Decanter	
Material	Plastic
Volume	2,900 gallons
Decant Time	480 minutes
Decant Split (Dense Phase)	
Methyl Ester	0.75% by Weight
Glycerol	95.00% by Weight
Methanol	70.00% by Weight
Soap	25.00% by Weight
Water	85.00% by Weight
Other	50.00% by Weight
Heat Loss	2.5%
Methanol Removal	
Methanol	98.00% by Weight
Water	8.00% by Weight
Temperature	180 °F

Heat Loss	2.5%	
Wash Tank		
Material	Plastic	
Volume	4,900	gallons
Wash Time	1,440	minutes
Wash Water Volume	50%	of Biodiesel Volume
Wash Steps	3	
Wash Losses		per Wash
Methyl Ester	0.50%	by Weight
Glycerol	98.00%	by Weight
Methanol	70.00%	by Weight
Soap	50.00%	by Weight
Water	55.00%	by Weight
Other	15.00%	by Weight
Heat Loss	2.5%	
Polishing Tank		
Material	Plastic	
Volume	4,900	gallons
Settling Time	1,440	minutes
Biodiesel Storage Tanks		
Material	Plastic	
Volume	5,000	gallons
Quantity	2	
Methanol Storage Tank		
Material	Steel	
Volume	6,000	gallons
Glycerin Storage Tank		
Material	Plastic	
Volume	2,500	gallons
Conversions		
Density	1.0	g/cc 8.3454 lbm/gal

Mass Balance

				Daily	Energy		
	Mol. Wt.	Density (g/cc)	Mass (lbm)	Volume (gal)	Cp (BTU/(lb-°F)	Change (BTU)	Temp. (°F) Comments
Pre-Heat							
<u>In</u>							
Yellow Grease		0.894	22,880	3,067	0.420		120
Triglycerides	868.66	0.894	21,278	2,852			
FFA	276.87	0.894	1,144	153			
Water		1.000	114	14			
Other		0.894	343	46			
<u>Out</u>							
Yellow Grease		0.894	22,880	3,065	0.420	192,188	140 Heat from Coil
Triglycerides	868.66	0.894	21,278	2,852			
FFA	276.87	0.894	1,144	153			
Water		1.000	114	14			
Other		0.894	343	46			
Reactor							
<u>In</u>							
Yellow Grease		0.894	22,880	3,065	0.420		140
Triglycerides	868.66	0.894	21,278	2,852			
FFA	276.87	0.894	1,144	153			
Water		1.000	114	14			
Other		0.894	343	46			
Methanol	32.04	0.791	4,051	613	0.623		55
Catalyst (KOH)	56.11	2.044	323				
<u>Out</u>							
Biodiesel		0.874	27,254	3,632	0.537	(37,380)	100 Loss to environment
Methyl Ester	290.90	0.874	21,313	2,922			
Glycerol	92.09	1.261	2,256	214			
Methanol	32.04	0.791	1,703	258			
Soap	314.96	1.000	1,371	164			
Water	18.02	1.000	218	26			
Other		1.000	393	47			
Decant							
<u>In</u>							
Crude Biodiesel		0.874	27,254	3,632	0.537		100
Methyl Ester	290.90	0.874	21,313	2,922			
Glycerol	92.09	1.261	2,256	214			
Methanol	32.04	0.791	1,703	258			
Soap	314.96	1.000	1,371	164			
Water	18.02	1.000	218	26			
Other		1.000	393	47			

Out

Biodiesel		0.874	23,034	3,139	0.537	(36,445)	94	Loss to environment
Methyl Ester	290.90	0.874	21,153	2,900				
Glycerol	92.09	1.261	113	11				
Methanol	32.04	0.791	511	77				
Soap	314.96	1.000	1,028	123				
Water	18.02	1.000	33	4				
Other		1.000	196	24				
Crude Glycerol			4,220	493	0.661		94	
Methyl Ester	290.90	0.874	160	22				
Glycerol	92.09	1.261	2,143	204				
Methanol	32.04	0.791	1,192	181				
Soap	314.96	1.000	343	41				
Water	18.02	1.000	185	22				
Other		1.000	196	24				

Methanol RecoveryIn

Crude Biodiesel		0.874	23,034	3,139	0.537		94	
Methyl Ester	290.90	0.874	21,153	2,900				
Glycerol	92.09	1.261	113	11				
Methanol	32.04	0.791	511	77				
Soap	314.96	1.000	1,028	123				
Water	18.02	1.000	33	4				
Other		1.000	196	24				

Out

Crude Biodiesel		0.874	22,531	3,063	0.537	1,335,946	180	
Methyl Ester	290.90	0.874	21,153	2,900		(54,445)		Loss to environment
Glycerol	92.09	1.261	113	11				
Methanol	32.04	0.791	10	2				
Soap	314.96	1.000	1,028	123				
Water	18.02	1.000	30	4				
Other		1.000	196	24				
Recovered Vapor								
Methanol	32.04	0.791	501	76		(212,332)		Heat of Vaporization
Water	18.02	1.000	3	0		(2,541)		Heat of Vaporization

WashIn

Crude Biodiesel		0.874	22,531	3,063	0.537		180	
Methyl Ester	290.90	0.874	21,153	2,900				
Glycerol	92.09	1.261	113	11				
Methanol	32.04	0.791	10	2				
Soap	314.96	1.000	1,028	123				
Water	18.02	1.000	30	4				
Other		1.000	196	24				
Water	18.02	1.000	36,496	4,373	0.989		55	

Citric Acid	192.12	1.665	313	23			
Out							
Biodiesel		0.874	21,089	2,887	0.537	(23,939)	85 Loss to environment
Methyl Ester	290.90	0.874	20,837	2,857			
Glycerol	92.09	1.261	0	0			
Methanol	32.04	0.791	0	0			
Soap	314.96	1.000	129	15			
Water	18.02	1.000	3	0			
Other		1.000	121	14			
Wash Water		1.000	38,251	4,586	0.989	(1,005,421)	85 Loss to Waste Water
Methyl Ester	290.90	0.874	316	43			
Glycerol	92.0938	1.261	113	11			
Methanol	32.04	0.791	10	2			
Soap	314.96	1.000	900	108			
Water	18.02	1.000	36,523	4,376			
Other		1.000	389	47			
Polishing							
Biodiesel		0.874	21,089	2,887	0.537		85
Methyl Ester	290.90	0.874	20,837	2,857			
Glycerol	92.09	1.261	0	0			
Methanol	32.04	0.791	0	0			
Soap	314.96	1.000	129	15			
Water	18.02	1.000	3	0			
Other		1.000	121	14			
Yield							
Theoretical	22,579	lbm					
Actual	20,837	lbm					
Yield	92.3%	by weight					
Batch							
Batch Size	800	gallons					
Batches per Day	4.60						
Time per Batch	3.48	hours					
Oil per Batch	667	gallons					
Methanol per Batch	133	gallons					
Pre-Heat Tank							
Retention	16.43	hours	Require 8 hours minimum				
Time to Temp	0.50	hours					
Reactor							
Retention	3.91	hours	Greater than Reaction Time	3.0		hours	
Bulk Capacity	192	gallons per hour	Less than Processor Rate of	200		gph	
Fill Time	40	minutes	Greater than	60		minutes	

Pelly A5

Retention 2.09 hours

Excess Capacity -67% Capacity less than 80 gpb

Decanter

Retention 12.78 hours

Batches to Fill 3.63

Average Decant Time 3.52 hours Less than 8.0 hours

Fill Time 145 minutes Greater than 60 minutes

Wash Tank

Retention 25.60 hours

Batches to Fill 6.13

Actual Batches Processed 4.60

Wash Time 24.00 hours Less than 24.0 hours

Fill Time 184 minutes Greater than 60 minutes

Polishing Tank

Retention 27.15 hours

Batches to Fill 6.13

Actual Batches Processed 4.60

Wash Time 24.00 hours Less than 24.0 hours

Fill Time 184 minutes Greater than 60 minutes

Biodiesel Storage

Retention 123.85 hours

5.2 days

Methanol Storage

Retention 267.90 hours

11.2 days

Glycerol Storage

Retention 121.73 hours

5.1 days

Transfer Times

60 minutes

Pump Capacity

20 gpm